

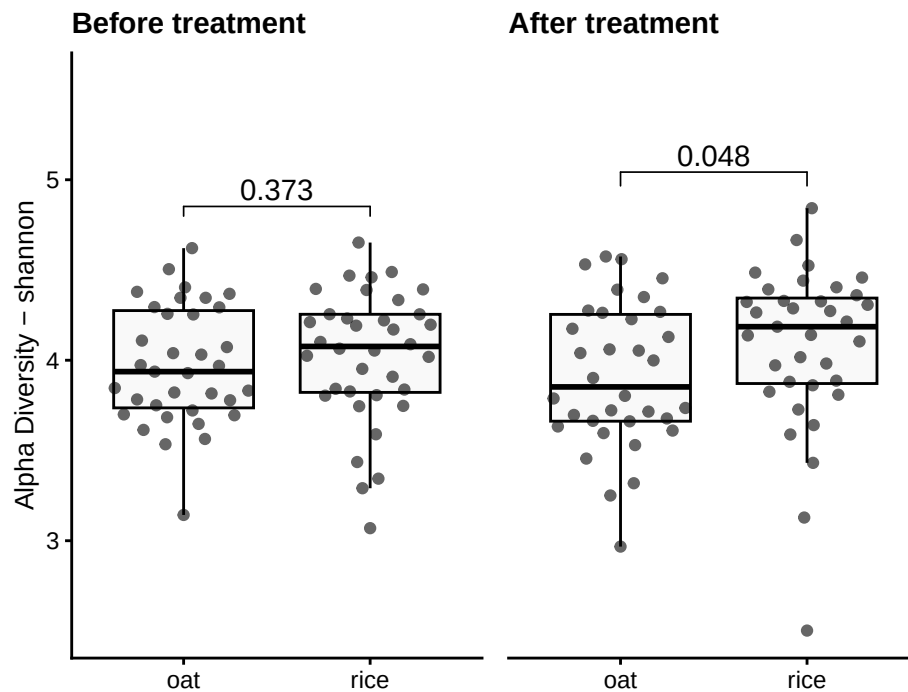
Alpha Diversity

Muluh

1 Community diversity

2 Group-wise comparisons

- **Index:** shannon
- **Group variable:** diet
- **Multiple testing correction:** FDR
- **Statistical test** Wilcoxon



2.1 Table of the top findings

2.1.1 Across-group

2.1.1.1 Before treatment

- Multiple testing correction: **FDR**
- Statistical test: **Wilcoxon**

Table 1: Table of alpha diversity comparisons before Treatment

group1	group2	mean1	mean2	log2FC	p.adj
oat	rice	4	4	0.02	0.373

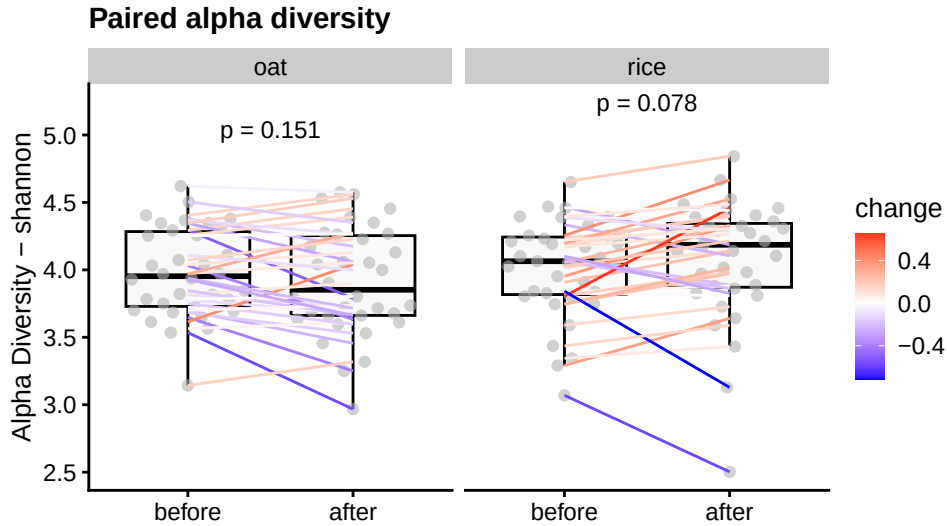
2.1.1.2 After treatment

- Multiple testing correction: **FDR**

- Statistical test: **Wilcoxon**

Table 2: Table of alpha diversity comparisons after Treatment

group1	group2	mean1	mean2	log2FC	p.adj
oat	rice	3.9	4.1	0.06	0.048



2.1.2 Within-group

2.1.2.1 Paired treatment

- Multiple testing correction: **paired** -none
- Statistical test: **Wilcoxon**

Table 3: Table of alpha diversity paired between timepoint(1 and 2)

diet	mean1	mean2	log2FC	p.value
oat	4	3.9	-0.02	0.151
rice	4	4.1	0.03	0.078

2.2 LMM

- Multiple testing correction: **FDR**
- Statistical test: **paired Wilcoxon**
- Formula used: $\sim \text{diet} * \text{timepoint} + (1 | \text{id})$

Table 4: Table of interaction effects

Term	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	4.035	0.089	135.874	45.491	0.000
dietrice	-0.076	0.125	135.874	-0.607	0.545
time	-0.062	0.043	67.447	-1.464	0.148
dietrice:time	0.127	0.060	67.442	2.119	0.038

3 Alternate analyses

- **Beta (Estimate)**: Shows the effect size and direction of a predictor on the outcome. Positive means an increase, negative a decrease.

Characteristic	before			after		
	Beta	95% CI ¹	p-value	Beta	95% CI ¹	p-value
(Intercept)	3.6	3.1, 4.0	<0.001	3.5	2.9, 4.1	<0.001
meal_group			0.3			0.3
1	—	—		—	—	
2	0.21	-0.03, 0.44	0.084	0.28	-0.02, 0.58	0.065
3	0.12	-0.12, 0.35	0.3	0.20	-0.09, 0.50	0.2
4	0.17	-0.06, 0.40	0.2	0.10	-0.20, 0.39	0.5
age	0.01	0.00, 0.01	0.2	0.01	0.00, 0.02	0.2

¹CI = Confidence Interval

Paired test between timepoints (1 and 2)	Beta	95% CI ¹	p-value
(Intercept)	3.5	3.1, 3.9	<0.001
meal_group			0.3
1	—	—	
2	0.23	-0.07, 0.53	0.14
3	0.16	-0.14, 0.46	0.3
4	0.13	-0.05, 0.31	0.2
age	0.01	0.00, 0.01	0.039

¹CI = Confidence Interval

- **95% CI:** Range where the true beta is likely to fall, reflecting precision.
- **p-Value:** Tests if beta differs from zero; $p < 0.05$ suggests significance.

4 Data Summary

```

class: TreeSummarizedExperiment
dim: 1700 140
metadata(0):
assays(1): relabundance
rownames(1700): SGB4933 SGB4285 ... SGB15018 SGB4921
rowData names(9): kingdom phylum ... strain clade_name
colnames(140): AE1332 AE2332 ... VK1336 VK2336
colData names(15): sample id ... shannon group
reducedDimNames(0):
mainExpName: NULL
altExpNames(20): kingdom phylum ... KO metacyc
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL

```